

Ishakya Gamage

No. 12, Sisilasa Uyana, Kamburugamuwa, Matara.

Phone: +94 71 147 5700 | Email: ranhiruugi.23@cse.mrt.ac.lk

ishakya.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of Moratuwa

BSc Engineering Honours in Computer Science & Engineering

Moratuwa, Sri Lanka

2024 – Present

- **CGPA:** 3.18 / 4.00

Rahula College

Primary & Secondary Education

Matara, Sri Lanka

2010 – 2023

- **Leadership:** Treasurer of the Media Unit (2021–2022) and Junior Prefect (2012–2013).

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

Frameworks & Libraries: Node.js, React, Next.js, FastAPI, Express.js, Nestjs

Tools & Cloud: NGINX, Redis, Vercel, Render, GCP, MySQL, PostgreSQL, Git, Ubuntu Linux

Other Concepts: Machine Learning, Multi Agentic Systems, DSP, Hardware/IoT (ESP32)

SOFTWARE ENGINEERING PROJECTS

SoundScout AI | *Multi-Agent System, WhatsApp Integration, Vercel*

[GitHub](#) | [Live App](#)

- Engineered an intelligent AV logistics marketplace and multi-agent system to automate event logistics and vendor bidding, submitting the prototype for IDEALIZE 2026.
- Built a four-repository architecture (frontend, backend, AI engine, WhatsApp worker) deployed on Vercel, featuring instant rentals with SSE-based inventory updates.

Sonora AI | *FastAPI, Next.js, DSP, Machine Learning, GCP*

[GitHub](#) | [Live App](#)

- Architected a full-stack AI-powered vocal production platform using Python/FastAPI and TypeScript/Next.js, featuring microtonal pitch correction and neural source separation.
- Integrated Google Gemini to dynamically generate timestamped automation maps for DSP mastering chains, enabling intelligent, genre-aware audio processing.
- Implemented enterprise-grade security using OAuth2/OIDC with PKCE via WSO2 Asgardeo, securing the application with a Backend-for-Frontend (BFF) pattern and RFC 7662 token introspection.
- Managed production infrastructure by deploying the backend on a GCP Compute Engine VM with systemd and Caddy TLS termination, alongside a Vercel-hosted frontend.

Another Home - Hostel Management System | *Node.js, MySQL, WSO2, REST API*

[GitHub](#)

- Developed backend infrastructure for a comprehensive hostel management platform, including global API versioning and API contract integrations.
- Scaffolded a dedicated authentication microservice managing a secure MySQL user schema for distinct warden and student roles.
- Implemented enterprise-grade security and identity management using OAuth2/OIDC via WSO2 Asgardeo.

SwiftDrop Flash Sales Platform | *NGINX, Node.js, Redis, Machine Learning*

[GitHub](#)

- Engineered a high-concurrency delivery platform utilizing multi-node NGINX load balancing to seamlessly handle severe flash sale traffic spikes.
- Integrated machine learning-based bot detection algorithms and secure JWT authentication to secure rapid transactions.

PredictiveOps - SE Project | *Go, Python, Kafka, Machine Learning*

[GitHub](#)

- Engineered an event-driven microservices architecture using Go and Python, leveraging Kafka for high-throughput, zero-data-loss telemetry stream processing.
- Developed an end-to-end Machine Learning pipeline transitioning from unsupervised Isolation Forest anomaly detection to supervised XGBoost models.
- Implemented a human-in-the-loop validation system and deployed production models via FastAPI, managed through MLflow with PostgreSQL for time-series persistence.

HACKATHONS & COMPETITIONS

- **2nd Runners-Up | BitCode v6.0:** Competed as Team Algostrom (UoM) in the national-level coding contest organized by Rajarata University.
- **National Finalist | OctWave 2.0 Kaggle Challenge:** Reached the Top 8 (Team OW-74 Byte Squad) in the advanced AI/ML data science challenge organized by IEEE IAS UoM.
- **Grand Finale Finalist | Sri Lankan Robotics Challenge (SLRC) 2025:** Selected to compete in the ultimate robotics showdown arena organized by ENTIC UoM.
- **Global Competitor | IEEEExtreme 19.0:** Competed in the global 24-hour algorithmic coding competition.
- **Active Prototype & Proposal Submissions (2026):** Successfully passed initial phases and submitted prototypes/proposals for *IDEALIZE 2026* (AIESEC), *LAUNCH'26* (IEEE CS UoK), *DHack'26* (USJ), *AI Challenge Sri Lanka* (IEEE YP), *GENESIZ '26* (KDU), *CodeSplash '26* (UoK), and *CodeRally 7.0* (IIT).

LEADERSHIP & VOLUNTEER EXPERIENCE

Department Representative (CSE Batch '23)

Department of Computer Science & Engineering

University of Moratuwa

Jan 2026 – Present

- Acting as the core liaison between the student body and faculty administration across semesters 4, 5, and 6 to coordinate academic schedules, deadlines, and batch-wide student welfare initiatives.

Event Logistics Lead & Co-Lead

IEEE Student Branch & Major University Events

University of Moratuwa

2025 – 2026

- **IEEE Student Branch:** Currently co-leading the permanent logistics committee, supervising inventory, vendor databases, and operational frameworks while mentoring junior volunteer committees.
- **MoraXtreme 10.0 & IEEE Open Week 2026:** Spearheaded 24-hour physical infrastructure for the flagship hackathon, and directed overall event operations for Open Week including massive 72-foot visual displays.
- **Hit the Grounds 2025:** Co-led end-to-end logistics and assisted with publicity for the flagship orientation.

Organizing & Delegate Handling Committees

CSESS, SLIoT, AIESEC, IESL

Multiple Organizations

2024 – 2026

- **Delegate & Corporate Handling:** Managed national-level contestants and VIP judges for *SLIoT 2026*, and coordinated top-tier tech company booths as a corporate liaison for *CSE Careers Day 2025*.
- **Event Execution:** Contributed to operational planning and live stream setups for *RoboGames 2026*, planned corporate networking sessions for *CSESS*, and managed marketing/delegate tracking for AIESEC's *On the Map 3.0*.

REFERENCES

Prof. Dulani Meedeniya

Department of CSE

University of Moratuwa

dulanim@cse.mrt.ac.lk

5th Semester Coordinator

Dr. Kutila Gunasekera

Department of CSE

University of Moratuwa

kutila@cse.mrt.ac.lk

PhD (Monash), BSc Eng. (Hons)